OCTO-EARTH

Integrated Project Report

Orbital relocation concept, launcher and power-system calculations, climate mirrors, and reinforcement-learning evaluation

EXECUTIVE SUMMARY
OCTO-EARTH is a hypothetical simulation in which an eight-arm launcher ejects asteroid material so the Earth–Moon system recoils from a 1 AU orbit toward 2.7 AU. A solar-collector belt beams power to the launcher, while a space-mirror system supplies additional sunlight. A PPO controller selects launch and climate actions. The submitted evidence supports a functioning simulation and one successful saved-checkpoint test; it does not establish real-world engineering feasibility, climate safety, or robust reinforcement-learning performance.

Project objective

The model targets a semi-major axis of 2.70 AU (±0.05 AU), eccentricity ≤0.05, inclination ≤5°, and a 10,000-simulated-year deadline. It also tracks Moon retention, Earth–Mars separation, a 288 K climate target (±5 K), and finite mass and energy budgets. These are model acceptance criteria rather than a proposal to move Earth in reality.

System at a glance

Subsystem	Modelled role	Key declared scale
E1 — launcher	Ejects reaction mass; Earth recoils by momentum conservation.	8 arms; up to 10¹■ kg per group packet; up to 5,000 km/s; 8 launches/year; 50% kinetic efficiency.
E2 — solar power belt	Collects sunlight and beams microwave power to arm-tip rectennas.	5 million collectors at 0.5 AU; 30% of solar output intercepted in the model; 5.8 GHz beams.
E3 — mirror ring	Reflects additional sunlight onto Earth at 2.7 AU.	1.6×10¹■ m² film area; 90% reflectivity; terminator orbit at 20 Earth radii.
PPO controller	Chooses launch throttle, direction, speed and mirror-light set-point.	31-value observation; 5 continuous actions; decisions every 100 years.

How to read this report

Pages 2–4 summarize the physics, launcher structure and thermal loads, and solar power / mirror calculations. Pages 5–7 describe the simulation architecture, reinforcement-learning evidence, and comparative mission outcomes. Page 8 records limitations and the recommended validation work. Values are attributed to the supplied reports; corrections or caveats are explicitly identified rather than silently merged.

1. Mission physics and order-of-magnitude calculations

The concept relies on conservation of momentum. When a packet of asteroid material is ejected in one direction, the remaining Earth–Moon system recoils in the opposite direction. Repeated, carefully directed impulses can change orbital energy and angular momentum. The model assumes the asteroid is docked to Earth and treats the Earth, Moon and remaining reservoir as a combined system for its primary estimates.

1.1 Mission constants and targets

Quantity	Value used	Meaning
Solar gravitational parameter, μ	1.32713×10^{2}■ m^3/s^2	Approximate Sun–system orbital parameter.
Initial / target orbit	1 AU → 2.7 AU	Target distance from the Sun.
Earth–Moon–asteroid system mass	6.0756×10^{2}■ kg	Model mass for the orbital screen.
Mission horizon / control interval	10,000 years / 100 years	At most 100 policy decisions.
Asteroid reservoir	3.00×10^{22} kg	About 0.408 Moon masses and 0.502% of Earth mass.

1.2 Circular-orbit speed screen

For a circular orbit around the Sun, the speed is $v = \sqrt{(\mu/a)}$. Using $a_{■}$ = 1 AU = 1.496×10^{11} m gives $v_{■} \approx$ 29,784.7 m/s. At $a_{■}$ = 2.7 AU = 4.039×10^{11} m, $v_{■} \approx$ 18,126.4 m/s. The supplied calculation therefore uses a slow-spiral speed-change screen of $\Delta v = v_{■} - v_{■} \approx$ **11,658 m/s**. A Hohmann-transfer reference is approximately 10,997 m/s, but its two large burns do not represent the gradual launcher strategy.

1.3 Orbital energy and power

The change in circular-orbit energy is estimated by $\Delta E = \mu M (1/(2a_{■}) - 1/(2a_{■}))$, giving about **$1.70\times10^{33}$ J**. Spread over 10,000 years ($T \approx 3.16\times10^{11}$ s), the ideal average orbital-energy rate is $P = \Delta E/T \approx$ **5.38×10^{21} W**. This is roughly 30,982 times the sunlight currently intercepted by Earth, emphasizing the extreme scale.

1.4 Launcher mass, speed and energy

The model sets a maximum packet mass of 10^{1}■ kg, a maximum launch speed of 5,000 km/s, and eight group launches per year. With launch efficiency $\eta = 0.50$, the minimum-energy screen uses the entire 3.00×10^{22} kg reservoir at a launch speed around 2.355×10■ m/s. The estimated launch energy is **1.66×10^{3}■ J**, averaging 5.27×10^{23} W over 10,000 years. At the 5,000 km/s cap, the model estimates a minimum transfer time of 1,771 years under maximum packet rate and speed.

Screening metric	Reported value	Interpretation
Earth escape speed with docked asteroid	≈11.2 km/s	Negligible relative to the assumed thousands of km/s launcher speed.
Packet kinetic energy at 2,480 km/s	$\approx3.07\times10^{3}$■ J	Per 10^{1}■ kg group packet.
Power drawn per packet at 50% efficiency	$\approx6.15\times10^{3}$■ J	About half becomes packet kinetic energy; the remainder is loss/heat in the model.
Peak launcher power at 5,000 km/s	$\approx6.34\times10^{2}$■ W	Exceeds ordinary planetary-scale energy infrastructure by an enormous margin.

Interpretation: “Feasible” in the calculation report means the chosen idealized mass, speed, energy and timing constraints pass a model gate. It does not mean the launcher, asteroid docking, power network or Earth-safe operation can be built.

2. Eight-arm launcher: structural and thermal screening

The supplied calculation insert assumes a packet follows a curved guide with radius R■ = 3R■. The guide's centripetal force is F = mv²/R■. It assumes eight equal load paths and an ideal circular load-bearing cross-section for each arm. These assumptions are critical; the numbers are not a completed structural design.

2.1 Static stress equation and radius ambiguity

With total load shared equally across eight arms, the assumed total allowable force is F■■■■w = 8πr■■■²σ■■■■w. Equating this to mv²/(3R■) gives the supplied speed limit:

v■■■ = √(24π σ■■■■w r■■■² R■ / m)

Using σ■■■■w = 86.67 GPa, r■■■ = 3.0×10■ m, m = 10¹■ kg and R■ = 6.371×10■ m gives v■■■ ≈ **1.9357×10■ m/s (19,357 km/s)**. The stated 86.67 GPa allowable corresponds to a 130 GPa ultimate strength divided by a safety factor of 1.5. However, the report itself warns that if "3,000 km" is the arm reach/length rather than the cross-section radius, substituting it for r■■■ is invalid.

Case	Total radial load	Per-arm load (equal split)	Mean stress / assumed area
Operating cap: 5,000 km/s	1.308×10²■ N	1.635×10²³ N	5.78 GPa (6.67% of allowable)
Equation limit: 19,357 km/s	1.960×10²■ N	2.451×10²■ N	86.67 GPa (100% of allowable)

Stress scales with v² under this simplified model. The 5,000 km/s operating speed is about 25.8% of the calculated limit, which yields 6.67% of the assumed allowable stress. This margin applies only to the stated idealized load and geometry.

2.2 Waste heat per launch and across the mission

Quantity	Calculation / assumption	Reported estimate
Packet kinetic energy	½mv², m=10¹■ kg, v=5×10■ m/s	1.25×10³¹ J
Energy drawn at 50% efficiency	Ekin/η	2.50×10³¹ J
Waste heat per event	Edraw – Ekin	1.25×10³¹ J
Per arm per event	Equal split across 8 arms	1.56×10³■ J
Average heat at 8 events/year	Annualized, not pulse power	3.17×10²■ W total; 3.96×10²³ W/arm

The heat figure is a system-level loss estimate. It does not mean all waste heat is deposited directly into the arm: power electronics, rectennas, conductors and radiators may receive different shares. If fraction f■■■ of one arm's event heat is deposited locally over pulse duration t■, a first pulse-power estimate is P■■■■■ = f■■■(1.56×10³■ J)/t■. Neither f■■■ nor t■ is supplied, so the actual temperature rise cannot be computed from the reports.

2.3 Thermal stress and combined failure checks

For a uniformly heated member whose expansion is fully restrained, σ■■ ≈ EαΔT. A freely expanding member has much lower uniform axial thermal stress, but gradients, constraints, joints and rapid heating/cooling can cause bending, local stress and fatigue. The supplied illustrative example (not a specified arm material) uses E=200 GPa and α=12×10■■/K, giving 2.4 MPa/K; ΔT=100 K implies 0.24 GPa and ΔT=500 K implies 1.20 GPa.

Required before a safety claim: specify actual cross-sectional area and curvature, material strength versus temperature, elastic modulus, thermal expansion, conductivity, heat capacity, emissivity, heat-deposition fraction, pulse duration, radiator geometry, support/joint loads, buckling, stress concentrations, fatigue and creep. Combine mechanical and thermal stresses at the worst point and time.

3. Solar power belt, microwave transfer and climate mirrors

The energy chain is sunlight → thin-film collectors → electrical power → 5.8 GHz microwave transmission → rectennas at the eight arm tips → launcher. The mirror ring is a separate climate subsystem that reflects sunlight onto Earth when its orbit is farther from the Sun.

3.1 Collector belt and microwave chain

Parameter	Model value / result	Explanation
Collector belt	5 million collectors at 0.5 AU	Assumed to intercept 30% of solar output; construction and station keeping are idealized.
Sunlight intercepted	≈1.15×10²■ W	Model's 30% solar-output assumption.
Microwave transmitter efficiency	80%	The report's energy ledger uses 20% transmitter loss.
Capture fraction	99.88% at 1 AU; 86.30% at 2.7 AU	Calculated belt geometry and blockage assumptions.
Rectenna diameter / count	1,000 km / 8	Each has area $\pi(5\times10■\ \text{m})^2 \approx 7.85\times10^{11}\ \text{m}^2$.
Rectenna conversion efficiency	85%	Delivered electricity is 1.29×10²■ W at 1 AU and 1.11×10²■ W at 2.7 AU.

At the target orbit, the modeled electrical output is reported as about 1.76 times the maximum launcher peak demand. This is an idealized aggregate power comparison, not proof that power can be routed, phased, received and dissipated safely at the required scale. The report estimates 5.30×10³■ J transmitter heat and 5.48×10³³ J of beam spill over the tested heuristic mission.

The calculations report a microwave photon-momentum side effect of 114 m/s using $\Delta v = E/(cM)$, but the calculation review flags this as not applied to the simulated dynamics. Momentum transfer from absorbed radiation should be accounted for consistently in the physical model; the stated figure should not be treated as an implemented Earth velocity change.

3.2 Collector station keeping and manufacturing

The belt assumes sail-tilt station keeping and factories using Mercury-derived material. In the calculation report, a 3,600-year run with station keeping has maximum slot deviation of 723 km, whereas the no-station-keeping comparison drifts to 1.46×10■ km and reaches eccentricity 0.127. The updated PPO report separately summarizes approximately 386 km maximum deviation, RMS 317 km, and roughly 51 m/s/year control Δv. These figures are from different run summaries and should not be merged as if they were one identical run.

3.3 Mirror capacity and climate model

Quantity	Calculation	Result
Mirror film mass	1.6×10¹■ m² × 0.005 kg/m²	8.0×10¹² kg
Effective area at 45° tilt	$A_{eff}=A\cos45°$	1.13×10¹■ m²
Mirror capacity at 2.7 AU	$A_{eff} \times 0.90 \times 1361 / 2.7^2$	≈1.90×10¹■ W
Extra power needed to restore today's sunlight	$(1 - 1/2.7^2) \times 1.74\times10^{1}■$ W	≈1.50×10¹■ W
No-mirror equilibrium at 2.7 AU	Two-box energy-balance estimate	≈152 K; severe freeze in model

The climate model uses a surface layer coupled to a deep-ocean layer, with albedo increasing as the planet freezes. At 2.7 AU the model estimates only 13.72% of present sunlight naturally reaches Earth; the mirror capacity is intended to offset this. The mirror system reflects 90% of incident light, with the absorbed 10% radiated away. Mirror pointing, orbital dynamics, light distribution and real climate response are not demonstrated by the calculation.

Calculation-review caveat: the supplied review identifies a likely missing circular-aperture factor in the C121 beam-capture estimate (about 1.31% rather than 2.1% for the cited 100 km receiver case). This is a flagged calculation issue, not a resolved redesign.

4. Prototype architecture and how to reproduce the simulation

The README describes OCTO-EARTH as a Python reinforcement-learning prototype with a 3-D browser viewer. It is a visualization and simulation interface around a model; its rendered solar-system view is explicitly not to scale in the Overview mode.

4.1 Run procedure and interface

The README's run instructions are to install the dependencies with **pip install -r requirements.txt** and launch **python octoearth_rl.py**. The local viewer opens at **http://127.0.0.1:8800/** or the next free port. Training starts automatically; Ctrl+C stops it while retaining the saved model.

Viewer area	Purpose and outputs
Mission	Start/stop training; choose Live training, Best PPO model, Best hand-coded (heuristic), or No ejection; select Auto/CPU/GPU; inspect progress.
View	Switch Overview/Earth POV, camera tracking, visible layers and screen panels.
Params	Inspect parameters, units and tooltips; apply valid setting changes before starting training.
Results	Training curves, PPO/baseline comparisons, target pass/fail, decision table, Mars-safety table and resource counts. Rewards are shown separately from physical outcomes.
Guide / Ledger / Log	Parameter explanations; climate, energy, collector and enabler ledgers; runtime messages.

4.2 Visual modes and interpretation

Mode / key	What it shows
Overview	Sun, Earth, Moon, Mars, asteroid, trajectories and target. The README warns that this view is not to scale.
Earth POV / V	Close-up of Moon, asteroid, launcher ring and eight arms, magnetic field, solar wind, Hill sphere and Mars.
Power beams / B	Microwave beams at the arm tips; brightness follows recorded power.
Collector / C	Close-up of one solar-collector satellite.
Weather & life / W	Climate under space mirrors; "show without" displays the modeled no-mirror frozen state.
Space / R / L / Esc	Play-pause; reset; legend; close the open drawer. Drag to orbit and use the wheel to zoom.

4.3 Simulation/control architecture

The technical report describes a configuration and feasibility gate, a Gymnasium environment, a fast engine for PPO rollouts, a Mars safety shield, a higher-fidelity N-body engine for validation, a climate/energy ledger, and a reporting pipeline. The controller does not directly overwrite Earth's velocity: its action is decoded into packet mass, launch speed, direction in the radial–transverse–normal frame, and a climate-light set-point. The physics engine applies the reaction, charges resources, accounts for packet outcomes, advances Mars interactions and returns the next observation and reward.

Failed alternatives in the README: the viewer also exposes four separately modelled failed concepts—laser push, hydrogen harvesting, superconducting magnetic arms, and a Dyson beam on a reflector held by the arms. These are presented as comparative failure cases, not as working enablers.

5. Reinforcement-learning method and recorded training

The control problem is represented as a Markov decision process. One episode is a full mission. Each action is held over a 100-year simulated interval; the configured rate permits up to 800 packet-launch events in one step. The observation is a scaled 31-value vector, and the action is a five-value continuous vector.

5.1 Observation and action design

Observation group	Indices / count	Information
Earth–Moon barycentre	0–5 / 6	Position and velocity, scaled.
Orbit and resources	6–16 / 11	Semi-major axis, eccentricity, inclination, energy gap, Moon state, remaining mass/energy/time and capability estimates.
Previous action	17–21 / 5	Previous normalized control values.
Mars safety	22–26 / 5	Relative Mars position, distance to safe threshold and predicted closest approach.
Climate and mirrors	27–30 / 4	Surface/deep-ocean temperature error, sunlight share and mirror capacity.

Action	Meaning
a0	Ejection throttle; maps to 0–100% packet mass, with values below a 2% threshold producing no ejection.
a1, a2	In-plane and out-of-plane thrust direction in the rotating RTN frame.
a3	Launch speed mapped linearly from 20 km/s to 5,000 km/s.
a4	Climate-light set-point, requesting 0.5–1.5 times today's sunlight; mirrors add light up to capacity.

5.2 PPO configuration

Setting	Saved configuration
Algorithm / network	Proximal Policy Optimization (PPO), Stable-Baselines3; MLP [64, 64] for actor and critic; tanh activations.
Rollout / batch / epochs	1,000 / 200 / 10.
Learning rate	3×10■■.
Discount / GAE lambda	$\gamma = 0.995$; $\lambda = 0.95$.
Initial log standard deviation / entropy coefficient	–0.5 / 0.0.
Seed / training target	Seed 1; 1,000 episodes; deterministic evaluation every 50 episodes.

The reward combines progress toward target orbital energy and a terminal success bonus with penalties for energy, mass, time, eccentricity, inclination, Mars-safety proximity, climate deviation and terminal failure. The report notes a potential reward-objective problem: later policies conserve resources by applying almost no useful propulsion and miss the deadline, yet their reported rewards become less negative. The logs suggest policy collapse or objective misalignment, but do not isolate a single code defect.

5.3 What the training artifacts show

Evidence	Observed result	Interpretation
training_log.csv	1,000 stochastic episodes; 94,586 timesteps; 0 successful episodes.	Every recorded exploration episode failed.
Termination reasons	641 deadline; 359 resources_exhausted.	No stochastic training episode completed the mission.
evals.json	20 periodic deterministic evaluations; episode 50 succeeds; episodes 100–1,000 fail.	The early selected checkpoint's success did not persist in later evaluations.
Saved best checkpoint	Selected at episode 50, 3,790 timesteps; separate deterministic evaluation succeeds on seeds 1001–1002.	This is not the same experiment as the stochastic episode-50 training rollout, which failed.
Last checkpoint	At episode 1,000, final a≈1.00198 AU after 10,000 years; energy fraction 0.0017; success=0.	Not suitable as the demonstration model.

6. Mission results and policy comparison

The updated technical report compares no ejection, a hand-designed heuristic, and the saved best PPO checkpoint. The procedure reports fast-model results on three seeds, then one high-fidelity N-body mission for seed 2001 and a 200-year ballistic validation. A single high-fidelity seed is useful evidence but is not enough to establish generalization.

Policy	Fast success	High-fidelity result	Years	Final a / e	Mass used	Energy used
No ejection	0/3	Deadline; validation fails	10,000	1.0000 AU / 0.01593	0%	0 J
Hand-designed heuristic	3/3	Success; validation passes	3,600	2.7000 AU / 0.00889	95.25%	1.757×10³■ J
Best PPO	3/3	Success; validation passes	7,200	2.6972 AU / 0.00748	97.28%	1.839×10³■ J

6.1 Safety and climate

Metric	No ejection	Heuristic	Best PPO
Closest Mars pass	0.37211 AU	0.05005 AU	0.04928 AU
Configured safe distance	0.02456 AU	0.02456 AU	0.02456 AU
Mars-shield overrides	0	200	472
Final surface temperature	288.17 K	288.33 K	286.00 K
Maximum climate deviation	2.08 K	2.57 K	4.68 K
Years outside ±15 K livable band	0	0	0
Moon bound at mission end	Yes	Yes	Yes
200-year orbit validation	Fail: target not reached	Pass	Pass

The best PPO mission reaches a = 2.69719 AU, eccentricity 0.00748 and inclination 0.2616° after 7,200 years. Its 200-year coast test reports maximum |Δa| = 0.00282 AU and a bound Moon. The closest Earth–Mars distance is 0.04928 AU (about 7.37 million km), above the configured 0.02456 AU limit. However, the 472 shield overrides—2.36 times the heuristic's 200—mean that some safety performance comes from an external guardrail, not necessarily from the learned policy alone.

6.2 Resource and climate interpretation

Compared with the heuristic, PPO takes 3,600 additional years, uses about 4.7% more propulsion energy and about 2.0 percentage points more asteroid mass. The heuristic's maximum climate deviation is 2.57 K, leaving 2.43 K margin inside the ±5 K target band; PPO's 4.68 K deviation leaves only 0.32 K. Both are within the wider ±15 K band throughout the reported mission. Without mirrors, the model estimates severe cooling around 155–157 K at 2.7 AU; the simulation's habitability outcome therefore depends on the hypothetical mirror system.

Primary comparison conclusion: the saved PPO checkpoint succeeds in one reported high-fidelity mission, but the hand-designed heuristic is the stronger current baseline on time, resource use, climate margin and safety-shield interventions.

7. Limitations, calculation-review notes and next steps

The reports intentionally distinguish simulation consistency from physical buildability. The calculation review finds most elementary arithmetic consistent under the declared assumptions, but flags specific issues that must remain visible in any submission.

7.1 Calculation and evidence caveats

Issue	Why it matters / treatment in this report
Arm geometry and stress	The 19,357 km/s result assumes a 3,000 km cross-section radius and equal load sharing. If 3,000 km is arm length/reach, the calculation is not valid. No real structural or fatigue design is provided.
Thermal stress	The 50% launcher efficiency estimates lost energy, not actual arm temperature. Material, heat distribution, pulse duration and thermal properties are missing, so a definitive thermal stress cannot be calculated.
Impulse divided by initial mass (C39)	The calculation review calls this check invalid as a delivered Δv test for a changing-mass/recoil system. Retain 11,658 m/s as the ideal orbital screen, not as independently proven by that simple ratio.
Photon momentum (C81)	The calculation report lists 114 m/s as an indicative beam push but says it was not applied. A consistent momentum ledger is required before claiming full physical accounting.
C121 capture fraction	Review flags a likely missing circular-aperture factor: about 1.31% rather than 2.1% for the cited 100 km receiver case. Recompute before using this value.
Conflicting mission outcomes	The calculation-review PDF and updated RL report do not fully agree on PPO outcomes/override counts. This report uses the updated RL report for the PPO comparison and labels calculation-review values as review notes, not merged measurements.

7.2 Engineering and model limitations

The model assumes an asteroid reservoir of 3×10^{22} kg can be brought to and docked with Earth; packets of up to 10^{1}■ kg can be launched at up to 5,000 km/s; the launch system survives shock and does not damage Earth; collectors intercept 30% of solar output at 0.5 AU; microwave transfer can deliver enormous power; mirrors can be assembled and pointed; and radiators reject waste heat away from Earth. The reports explicitly identify hardware, launch shock, atmospheric effects, mirror orbit/pointing, light distribution, and real climate response as incompletely modelled or unmodelled.

7.3 Recommended next experiments

Priority	Action	Evidence required
1	Diagnose reward collapse; log each reward component on successful and failed evaluations.	Explain why near-zero propulsion can improve return; document reward changes against a hypothesis.
2	Preserve best checkpoint, final checkpoint and heuristic baselines separately.	Automated, versioned comparison on identical seeds.
3	Evaluate heuristic and PPO on 20–30 unseen randomized phases and multiple high-fidelity seeds.	Success rate and spread for time, mass, energy, climate, Mars distance and overrides.
4	Run safe ablations for the Mars shield, climate margins and collector availability.	Separate learned-policy performance from guardrail intervention and test sensitivity.
5	Recompute flagged beam capture and momentum terms; verify structural and thermal inputs.	Reviewed equations, named material/geometry, heat-flow assumptions and reproducible calculations.

Conclusion

OCTO-EARTH is a useful exploratory simulation integrating orbital mechanics, finite-resource propulsion, power beaming, climate control, visualization and PPO-based control. Under the model's declared hypothetical enablers, the selected PPO checkpoint passes one high-fidelity mission and a 200-year stability check. Nevertheless, the training log shows no successful stochastic episodes and all later periodic evaluations fail; the heuristic remains faster and more resource-efficient. The defensible conclusion is a simulation demonstration—not a feasible engineering plan for relocating Earth.

Source basis: supplied README(1).md, Calculations.pdf (calculation review), and OCTO_EARTH_RL_Report_Updated.pdf (run evidence dated 25 October 2026).